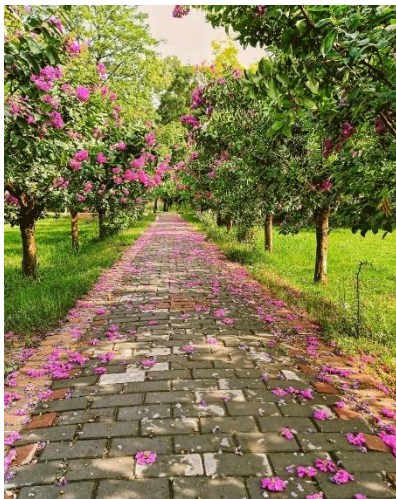


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ACADEMIC-UNDERGRADUATE STUDIES DIVISION
BITS F464- MACHINE LEARNING
COMPREHENSIVE EXAMINATION
SECOND SEMESTER 2019-2010

TIME: 180 MINUTES
DATE: 09.06.2020 / FN

MARKS: 40

PINK FLOWERED TREES (5 marks)



Assume the trees are data points. Find a suitable algorithm you would prefer to sketch this path?

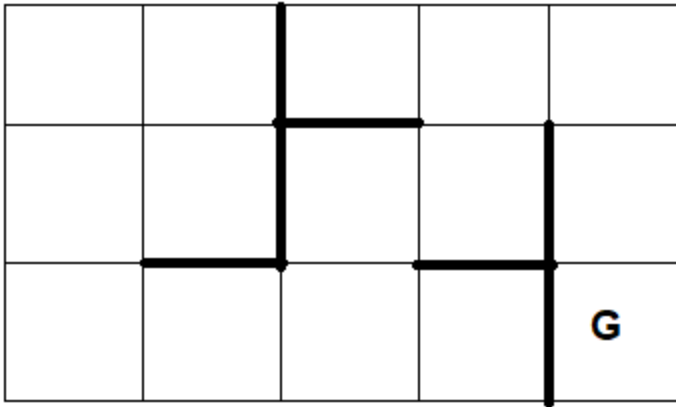
Draw a neat step by step derivation to find the width of the path separating pink flowered trees.

NOTE: STEPS OF DERIVATION IS MUST

.....
CAUGHT IN MAZE (5 marks)

Given the diagram below. You need to reach the destination. The highlighted lines represent the blocked path. The profit of reaching the goal state (G) is 100. Use a discounted reward of value 0.9. Apply the Q-learning ($Q(s, a)$), algorithm and calculate the in and out reward of moving from every state to every other state and finally reaching the goal state. (4 marks)

Also write the Q-learning reward calculation formulae. (1 marks)



NOTE: STEPS IS MUST. ANY MATH CALCULATION DO IT IN THE EXAM SHEET ITSELF.

BREAK INTO PIECES (5 marks)

Given below is the error equation of a single training point.

$$\text{Error}(y(x), t) = -t \log(y(x)) - (1-t) (1 - \log(y(x)))$$

$t \rightarrow$ target output

$y(x) \rightarrow$ predicted output

Guess the name of the model and draw its graphical representation. Write its math model equation. (2 marks)

The target t takes only binary values. Apply this over the error equation and reduce it. Draw its corresponding graphical representation. (3 marks)

REDUCE THE DIMENSION (5 marks)

Given 6 data points as $(1,1)$, $(2,1)$, $(2,0.7)$, $(2,3)$, $(2,4)$, $(2,3.5)$. The first three points belong to cluster 1 and the latter belong to cluster 2. Apply a supervised dimensionality reduction technique and find the relocated values of the points. (4 marks)

Did you find any change in the points allocated to clusters? (1 mark)

CRYING CHILD (5 Marks)

Scenario:

Design a directed acyclic graph, $G=\{v_1,v_2,v_3\}$ and $E=\{e_1,e_2\}$. Mother and father act as parent nodes, and they have one child that act as child node. The two directed edges connects the parent to their child node.

The child is dependent and will not stay alone in home. It depends on either father or mother to stay along with it else it begins to cry. Let's run a probability story around the child.

The probability that father being present in home is 0.8.

The probability that mother being present in home is 0.8.

The probability that the child is happy in the presence of both father and mother is 0.7.

The probability that the child is happy in the absence of both of them is 0.1.

The probability that the child is happy in the absence of mother but presence of father is 0.2

The probability that the child is happy in the absence of father but presence of mother is 0.2

Find the probability that the child cries when its father is absent.

STEP MARK APPLICABLE

PENALIZING THE ERR ONES (5 marks)

Write the penalty equation for polynomial regression algorithm. (2 mark)

How do you solve the problem of overfitting? (1 mark)

Rewrite the same penalty equation to overcome the overfitting problem. (2 mark)

SLIDE DOWN TO REACH NULL ERROR (5 marks)

Assume you apply delta rule over an unthresholded perceptron. To ease your job, I have mentioned the error equation below:

$$E(\vec{w}) \equiv \frac{1}{2} \sum_{d \in D} (t_d - o_d)^2$$

Where t is the target and o is the observed data values.

Derive the gradient descent algorithm and calculate the steepest gradient's direction. (3 marks)

Also calculate the weight update rule for gradient descent. (2 marks)

CATS AND DOGS (1 Mark)



Scenario: Given a dataset of cat and dog characteristics. You need to classify them rightly into cats or dogs?

You train and test the classification algorithm. The total accuracy you obtained during testing phase is 78%.

Which of the following is TRUE?

- a. 22% is considered as bias value.
- b. 22% is considered as variance.
- c. 22% is considered altogether for bias and variance during training and testing respectively
- d. Bias and Variance have nothing to do with dog and cat identification.
- e. Say the bias in training phase is p , then the variance in testing phase is $(1-p)$.

DID MY LOVER JOINED THE PARTY (2 Mark)

It's time to find that.

To know this, you get 5 audio clips from your friend who attended the party. You listen to these audio clips a number of times but still unable to identify your lovers voice from the played audio clips. So you run an algorithm to identify his/her voice. The details of the recorded audio clips is given below:

	Loud Music played behind	Large background voice	Many resembling voices
Clip 1	✓		✓
Clip 2	✓	✓	✓
Clip 3		✓	✓
Clip 4	✓		
Clip 5		✓	✓

- a. Guess the type of Machine Learning approach you would choose for this scenario
- b. Guess, would you have a high training error or high test error or both.

THREE HYPOTHESIS SPACE (2 Marks)

Before analyzing the dataset, a hypothesis is framed in general. In real time, you may encounter a number of hypothesis and not only one. You must ensure that you pick the right hypothesis as the proximity of the hypothesis towards the data set matters a lot.

Ex: Three Hypothesis space: You register ML subject with three different faculties and start learning the subject. The general likeliness probability of your seniors towards the three faculty's is given by 0.5, 0.3 and 0.2.

A new student comes in and attends all the three faculty classes. He raises a same True/False question to all the faculties. The answer is a binary answer {yes, no}.

If all the faculties answer TRUE then he confirms the answer to his question as TRUE. If all answer FALSE then he confirms the answer to his question as FALSE. If there is any chaos in their answer, the student is now in a position to finalize the right answer from all their answers given.

If the teacher 1 and teacher 3 answers to be true and teacher 3 answers it as false then find the probability whether the answer to the question is true or false. (2 marks)

HINT: Identify the classifier that you would prefer and then solve it.

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BITS F464- MACHINE LEARNING – TEST 2
SECOND SEMESTER 2019-2010

TIME: 50 MINUTES
DATE: 07.04.2020

MARKS: 20

Dear Students

Please do not share any contents with your friends related to exam during the exam duration.
Understand that we (Myself and you) put these efforts for the smooth conduction of the course.
Please be loyal to yourself.

1. Keep A4 sheets, pen, calculator and a gadget to scan the answer sheets and upload.
2. Do not close the browser for any reason.
3. Keep your BITS mail logged in.
4. Keep Impartus logged in.
5. Login in online tool with your BITS mail id.
6. Attend all the questions
7. No forward / Backward movement allowed in browser.
8. Test can be taken only once.
9. Upload everything and then submit within the time given.
10. The portals will close after the specific duration for sure.

Question pattern

PART A:

4 multiple choice question: Link shared in mail: Online Tool
MCQ portal will be open for first 15 minutes only.

QUESTION 1 (1 marks)
QUESTION 2 (2 marks)
QUESTION 3 (1 marks)
QUESTION 4 (2 marks)
Questions with multiple answers have internal mark split. Choosing the right combination will yield you full marks. Partial marks will be provided for partial answers.

NOTE: Do not take more than 15 minutes. Submit it at the earliest and move on to the next part.

PART B:

After 15 minutes a link will be shared for PART B

QUESTION 5 (4 marks)
QUESTION 6 (6 marks)
QUESTION 7 (4 marks)

NOTE: Do not take more than 35 minutes to solve. Additional 10 minutes to scan and upload the document. The link will be locked after 45 minutes.

Legibly solve them in A4 sheet and upload it as PDF.

1 PDF per question. i.e. 1 upload per question is allowed.

Answers in the link will only be evaluated. Do not send the answer scripts in mail which will not be evaluated.

QUESTION 1 (1 marks)

QUESTION 2 (2 marks)

QUESTION 3 (1 marks)

QUESTION 4 (2 marks)

QUESTION 5 (4 marks)

QUESTION 6 (6 marks)

QUESTION 7 (4 marks)

QUESTION 1 (1 marks)

Assume you are working with a two class (C1 and C2) classifier problem using Baye's classifier. Choose which of the following is true.

x - is the point that you are trying to classify.

$$P(C1/x)=P(C2/x)$$

$$P(C1/x)+P(C2/x)=1$$

$$P(x/C1)=1$$

$$P(x/C2)=1$$

$$P(C1/x)+P(C2/x)>1$$

QUESTION 2 (2 marks)

Let you run the Adaboost algorithm 3 times. The error in each stage is given to be $2/5$, $3/8$, $3/12$. Calculate the individual voting power of each of the hypothesis.

0.20273

0.235

0.549

1,1 and 1

0,0 and 1

0.2924

0.3681

0.79245

QUESTION 3 (1 marks)

Q1. The value of $k=1$ in KNN will lead to

Bias will be Zero, Perfect prediction of training data

New test data will not be predicted rightly

The variance may lead to infinity

Q2. The value of $K=N$ in KNN will lead to

Bias will increase, Bad prediction of training data

Variance may decrease

New test data will be predicted rightly

The model becomes complex.

QUESTION 4 (2 marks)

Let us assume a street has a Width (W_i) and the midline of the street has a equation of $8x-y-3=0$.

What is the width W_i , W vector and w_0 .

2.48, $[8, -1]$, -3

8.06, $[8, -1]$, 3

2.48, $[-8, 1]$, -3

0.030, $[-8, 1]$, -3

QUESTION 5 (4 marks)

Assume you are a scientist and fix four hypothesis $\{h_1, h_2, h_3, h_4\}$ to solve a problem. The probability of each of the hypothesis is given by $\{0.4, 0.2, 0.2, 0.2\}$.

The hypothesis 1 predicts that the problem would be solved

The hypothesis 2 predicts that the problem would be solved

The hypothesis 3 predicts that the problem would not be solved

The hypothesis 3 predicts that the problem would not be solved

Find whether the problem would be solved or not? **(1 mark)**

What is the probability? How do you solve this? **(2 mark)**

Choose the algorithm that would help in this situation? **(1 mark)**

QUESTION 6 (6 marks)

Given the two class of points $A=\{a_1=(1,0.5), a_2=(2,8), a_3=(4,0), a_4=(4,2)\}$ and $B=\{b_1=(6,2.5), b_2=(7,0.5)\}$. Identify the support vectors from the two class of points.

Find the coefficient of the dot product of the support vectors. **(5 mark)**

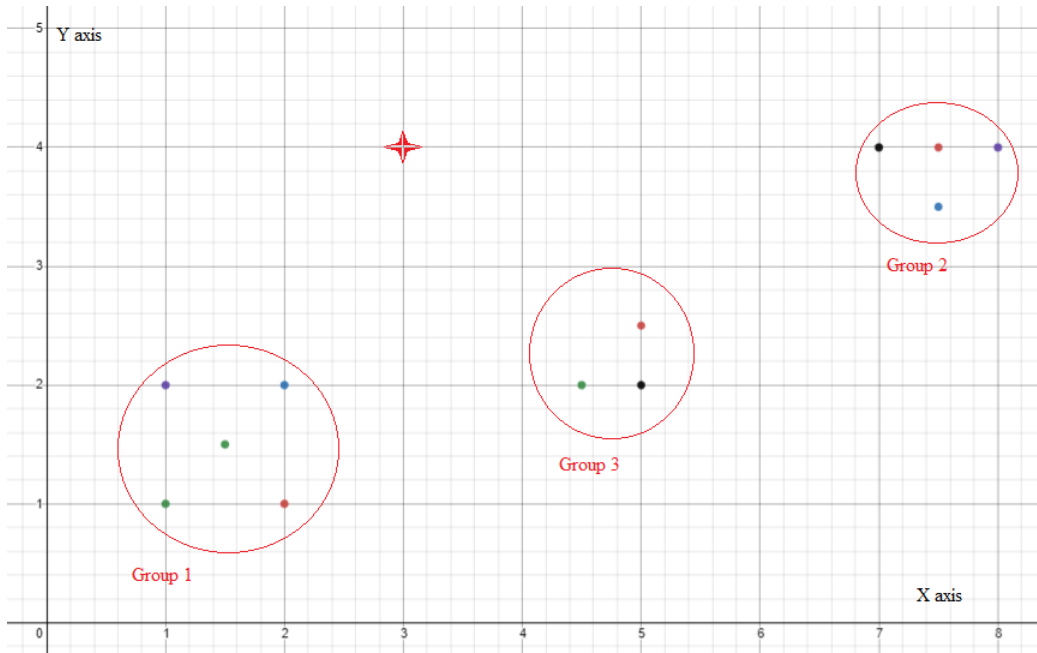
Also find the W vector. **(1 mark)**

(Step mark applicable)

QUESTION 7 (4 marks)

Use 4-NN classification and find to which category the “star point” belongs to. Get the co-ordinates of all the points from the graph

(Step mark applicable)



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ACADEMIC-UNDERGRADUATE STUDIES DIVISION
BITS F464- MACHINE LEARNING – TEST 1
SECOND SEMESTER 2019-2020

TIME: 50 MINUTES

MARKS: 20

DATE: 25.02.2020

Question	Points
1	0.5
2	0.5
3	1
4	4
5	4
6	5
7	5

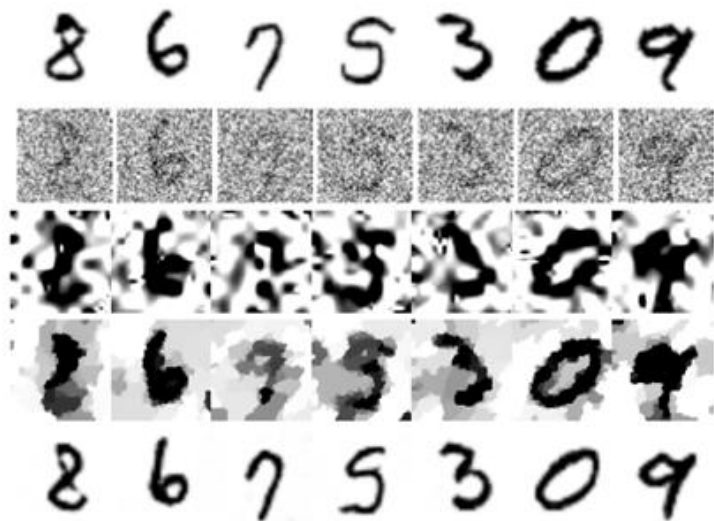
1. Analyze the picture given below.

What kind of machine learning algorithm can be applied in this scenario? (0.25 points)

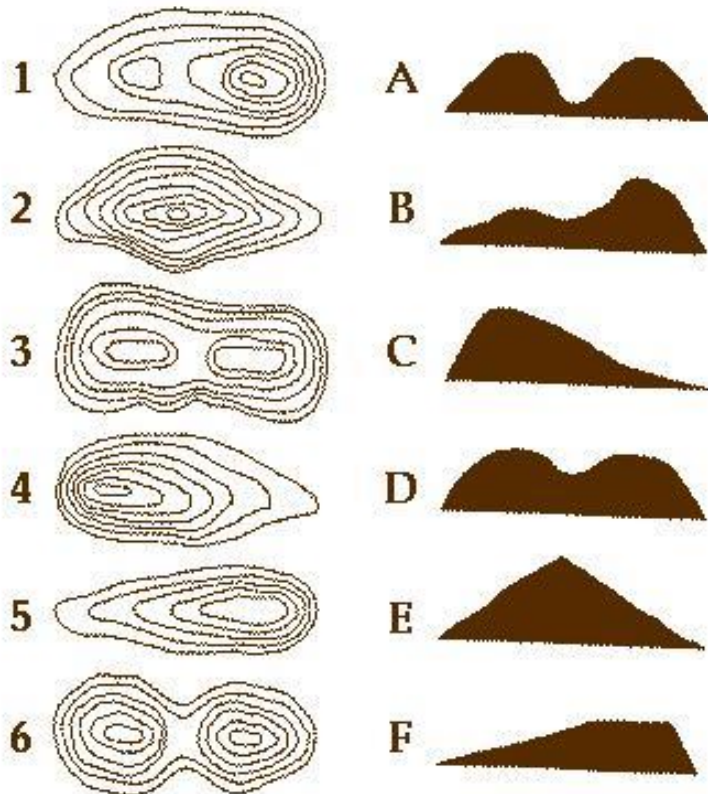
Justify with reasons. (0.25 points)



2. What kind of Supervised Learning models you could impose on this dataset to identify the alphabets. (0.5 points)



3. Map the right contour to the right gradient descent curve (1 point)



4. Let children's play with fruits. There are 2 types of fruit, each cut into many pieces. Now comes a question of classifying these pieces?? Which piece is from which fruit? You have a number of options left and few given below***:

1. You can guess from the color of the piece.
2. You can guess from the taste of the piece
3. You can guess from the shape of the piece.
4. You can smell the fruit piece and identify. Etc.

Mention the algorithm that you would prefer to classify them. (0.5 point)

Write its equation and explain its parameters. Detail the step by step solution. (3.5 points)

5. Given the data points (2,-1),(2,-2),(3,-1),(4,1),(5,1). Plot them in the graph given.

Assume you try two different regression equation which is given below

a. $x-y=4$

b. $x-y=3.5$

Find the suitable regression equation among both... (4 points)

6. Given a two class classifier problem and data points.

Constraints: The input data points range from $[-\infty$ to $+\infty]$. The size of the dataset is small enough.

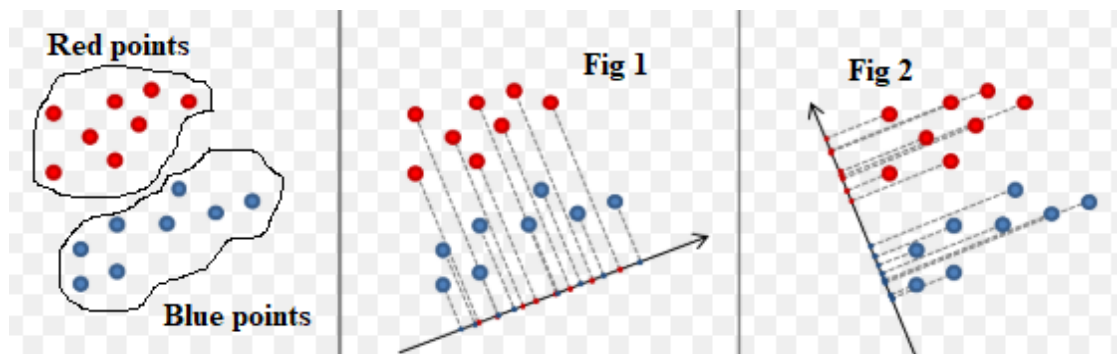
a. Identify the approach you adopt to predict the right class of each of the data point. (0.5 point)

b. Write a very short note on the approach chosen by you and derive the classifying boundary equation. (2.5 point)

c. Nevertheless, with any classifying equation, the error equation need to be framed for such problems. Identify the error equation too. (2 point)

NOTE: Use graphs wherever necessary

7. Given the data points in two different colors. The points are static in space. The points are projected onto two different vector lines as mentioned in Fig 1 and Fig 2 respectively.



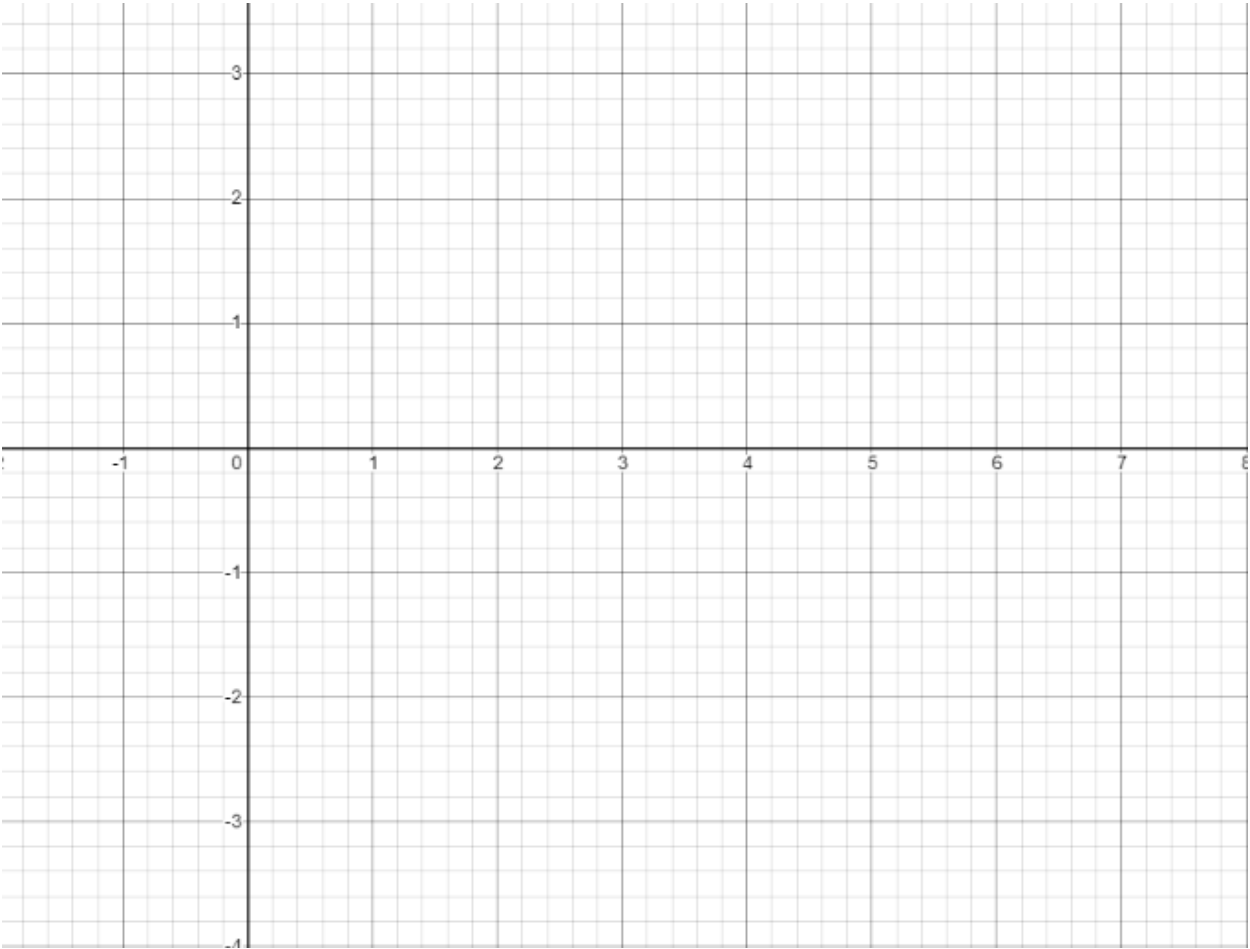
a. Which projection do you think is the finest separation? (0.5 point)

b. Identify the method you adopt and write its name (0.5 point)

c. Derive the vector equation over which the projected points are widely separated. (4 points)

NOTE: STEP BY STEP DERIVATION IS EXPECTED.

GRAPH SHEET



BITS PILANI DUBAI CAMPUS
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ACADEMIC-UNDERGRADUATE STUDIES DIVISION
BITS F464- MACHINE LEARNING – QUIZ QUESTIONS WITH SOLUTIONS
SECOND SEMESTER 2019-2020

TIME: 20 MINUTES

MARKS: 10

DATE: 08.03.2020

QUESTION 1

- Let H be the neuron in the hidden layer with logistic activation function. Input to the neuron would be H_I and output of the neuron be H_O . What would be the differentiated output of the H neuron when the neuron is executing.
 - ☒ $H_O * (1-H_O)$
 - ☐ $H_I * (1-H_I)$
 - ☐ $H_O * (1-H_I)$
 - ☐ $1/(1+e^{-(H_I)})$

QUESTION 2

- Identify the sentence that is TRUE with gradient descent.
 - ☐ Gradient will end in global minima always
 - ☐ Gradient specifies the rate of fastest decrease in Error
 - ☐ Gradient is a scalar quantity
 - ☒ Gradient specifies the rate of fastest increase in Error

QUESTION 3

- The prediction algorithm (Naïve Bayes) shows overfitting result. You try to rectify it by combining a number of different algorithms like (SVM, Regression, NN). Now choose which method you would adopt to combine the different algorithms.
 - ☒ Bagging
 - ☐ Boosting
 - ☐ Straight Linear Approach

QUESTION 4

- Which learner you would prefer.

Learner 1: 99% prediction false

Learner 2: 99% prediction true

Learner 3: 50% prediction true

- ☐ Learner 2 and Learner 3
- ☐ Learner 1
- ☐ Learner 2
- ☒ Learner 1 and 2

QUESTION 5

- Recollect Adaboost algorithm.

If the error in the classifier is $1/8$. What would be the voting power of the classifier.

- ☐ -0.97
- ☒ 0.97
- ☐ 1.0397

QUESTION 6

- What leads to overfitting

- ☐ high bias, low variance
- ☐ high bias, high variance
- ☒ low bias, high variance
- ☐ low bias, low variance

QUESTION 7

- Lets plot the following points in space.

A (2,1) positive

B (2,7) positive

C (4,3) positive

D (6,5) negative

E (8,7) negative

F (3,4) negative

Choose the best classifier with minimal error

☐ $(X \geq 4)$

☐ $(X > 3)$

☒ $(X \geq 6)$

QUESTION 8

- Assume you use a 4 input Linear Neural network with one hidden neuron (step activation function).

The 4 inputs are 1,2,10 and 100. The weight of the links are 0.1,2,3,-3. What would be the output of the Linear neural network.

☐ 1

☒ -1

☐ -infinity

☐ infinity

QUESTION 9

- Do you think ensemble models always improve the accuracy

☐ Yes

☒ No

QUESTION 10

- Which option is FALSE for 'momentum'
- ☐ It speeds up the process of gradient descent calculation
- ☐ It always depends on the step width you take
- ☐ Its weighted average of all the descents
- ☒ Always leads to global minima